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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 03 – Lecture 03: Dimensionality Reduction (PCA, LDA, ICA)

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Topic focus: **reduce dimensions** for speed, noise reduction, and visualization

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

Quick Links

Why Reduce? PCA LDA/ICA Example Summary

Agenda

Why reduce dimensions?

PCA

LDA and ICA (overview)

Mini example

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain why high-dimensional data can be difficult (curse of dimensionality).

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- Outline PCA steps and interpret explained variance.

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain why high-dimensional data can be difficult (curse of dimensionality).
- Describe PCA intuition (maximum variance directions).
- Outline PCA steps and interpret explained variance.
- Distinguish PCA vs LDA vs ICA at a conceptual level.

Abbreviations & Symbols

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- PC: principal component, EVR: explained variance ratio

Curse of Dimensionality (Intuition)

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- In high dimensions, data becomes sparse.
- Distances can become less informative.
- Models may overfit and require more data.

Dimensionality reduction helps with **compression**, **noise reduction**, and **visualization**.

PCA: Intuition

Principal Component Analysis (PCA)

Find orthogonal directions (components) that capture **maximum variance** in the data.

- First component: direction with highest variance.

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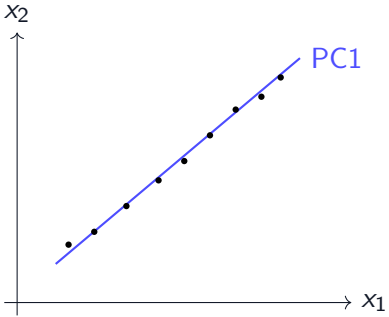
Find orthogonal directions (components) that capture **maximum variance** in the data.

- First component: direction with highest variance.
- Next components: highest remaining variance, orthogonal to previous.

PCA Geometry (2D Intuition)



PCA chooses the direction with maximum spread (variance) and projects data onto it.

PCA Steps (High Level)

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4. Choose top-k eigenvectors (components).
5. Project data onto these components.

Key Formulas (What PCA computes)

- Centered data: $X_c = X - \mu$

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- Covariance: $C = \frac{1}{n-1} X_c^\top X_c$
- Eigen-decomposition: $Cv_i = \lambda_i v_i$
- Projection (keep top-k): $Z = X_c V_k$

V_k contains the top-k eigenvectors (principal directions).

Explained Variance

- Eigenvalues indicate how much variance each component captures.

Example rule: choose the smallest k with $\geq 90\%$ explained variance (depends on use-case).

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Explained Variance

- Eigenvalues indicate how much variance each component captures.
- Explained variance ratio helps choose k.
- Goal: keep enough components to retain most information.

Example rule: choose the smallest k with $\geq 90\%$ explained variance (depends on use-case).

Choosing k (Scree Plot Idea)

- Plot EVR per component: large drop $\Rightarrow$ "elbow" point.

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- Plot EVR per component: large drop $\Rightarrow$ "elbow" point.
- Or use cumulative EVR: pick smallest k reaching a target (e.g., 90%).
- Trade-off: compression vs information loss.

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- **PCA**: unsupervised; maximizes variance.
- **LDA**: supervised; finds projections that separate classes.
- **ICA**: unsupervised; finds statistically independent components (source separation idea).

Quick Comparison Table

Method	Uses labels?	Goal	Typical use
PCA	No	maximize variance	compression/visualization
LDA	Yes	separate classes	classification features
ICA	No	independent sources	signal separation

Mini Example (2D to 1D Projection)

- Suppose points lie roughly along a diagonal line in 2D.

In the lab, you will implement PCA and visualize explained variance.

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Mini Example (2D to 1D Projection)

- Suppose points lie roughly along a diagonal line in 2D.
- PCA finds the diagonal direction and projects data to 1D.
- Result: compressed representation with minimal information loss.

In the lab, you will implement PCA and visualize explained variance.

Minimal PCA in Python (sketch)

```python
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

X_scaled = StandardScaler().fit_transform(X)
pca = PCA(n_components=2)
Z = pca.fit_transform(X_scaled)
print(pca.explained_variance_ratio_)
```

Standardize first; then inspect explained variance ratios.

Summary

- Dimensionality reduction combats high-dimensional issues and improves efficiency.
- PCA finds maximum-variance orthogonal directions; explained variance guides k.
- LDA uses labels to separate classes; ICA seeks independent components.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

Exit Questions

1. What does PCA optimize?
2. Why do we often standardize before PCA?
3. Compare PCA and LDA in one paragraph.
4. What does "explained variance" mean?